Most authorization systems use these four elements, either explicitly or implicitly, when evaluating policies to make access decisions. They enable policy languages to express rich, contextual rules while keeping complex policies understandable and maintainable.

4.3.2 Understanding the language of access

Now that we've established the PARC model, let's take a closer look at the vocabulary commonly used in access control and authorization. These terms appear in policy languages, IAM platforms, and system documentation, but they're not always used consistently. In this section, we'll define them precisely and explain how they relate to one another, and to the decisions that systems make about access. A clear understanding of these terms will help you design better policies, reason about authorization systems, and communicate more effectively across technical and business roles.

The following is a list of words, in addition to those defined in the last section, that we'll use frequently in our discussion of authorization.

- **Entity**—a structured object within the authorization system that participates in policies. Entities include principals such as users, devices, or services, as well as resources like documents, records, or applications. Each entity can have attributes that influence policy decisions.
- **Subject**—in chapter 2, we defined *subject* as the "entity being referred to in an account, policy, or decision." In this book, I will avoid using the word *subject* to minimize confusion, as it could refer to either the principal or the resource depending on the context. Some historical systems use subject instead of principal, and I will highlight those instances when necessary.
- **Group**—a collection of entities that can be referenced collectively in a policy. Groups simplify access management by allowing shared rules to be applied to multiple users or resources simultaneously. For example, all members of the `Customer-CustCo-Users` group might be allowed to view documents owned by CustCo.
- **Namespace**—a prefix that organizes and clarifies entities and actions. Namespaces prevent collisions and help specify to which domain or